

“PAPER_{0:D3}” -f [int]# PAPER #74 □ Galactic Structure: NED + SIMBAD + UQFF Cross-Validation

Title: Galactic Structure Cross-Validation: NED and SIMBAD Multi-Object Queries vs UQFF Predicted Velocity Dispersions and Mass Profiles

Author: Daniel T. Murphy

Framework: UQFF Star-Magic (? = 0.0005/day, [SSq] = 0.57)

Date: March 7, 2026

Source Data: QCalc_validation.py (DataSourceURLs: NED_BASE, SIMBAD_BASE, SIMBAD_API)

Index Slot: §1.10 Database Integration & Multi-Wavelength Astrophysics,
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1. Database Query Architecture

NED TAP ADQL Query (Galaxy Velocity Dispersions)

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AND vel_disp IS NOT NULL
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$$\sigma_{\text{UQFF}}^2 = \sigma_{\text{Newton}}^2 \times \left(1 + \frac{F_{U,Bi,i}}{F_{\text{Newton}}} \right) = \frac{GM}{r_{\text{eff}}} \times \left(1 + \frac{\Omega_g \cdot \sum U g_j}{GM/r^2} \right)$$

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Average UQFF enhancement: s_UQFF/s_Newton = **1.018** (= [SSq] × 0.032 correction factor).

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For M31 (Andromeda): $r_{\text{AGN}}/r_{\text{gal}} \sim 0.001$, so $d\mu \sim 0.057\%$ within SIMBAD proper motion uncertainties ($> 10\%$) for extragalactic objects.

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When SIMBAD + NED + GAIA data are combined for the same galaxy:

Parameter	SIMBAD	NED	GAIA	UQFF
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Proper motion	□	□	?	+ $d\mu$ (negligible)

Summary

Database	Query Method	UQFF Prediction	Agreement
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When SIMBAD + NED + GAIA data are combined for the same galaxy:

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Photometric M_star	$\square$	?	?	Input
Proper motion	$\square$	$\square$	?	+d μ (negligible)

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UQFF computed: Galactic scale UQFF gravity correction $g_{\text{UQFF}}/g_{\text{Newton}} = 1 + [SSq] \times ? \times (r/\text{kpc})^2 = 1 + 2.85\text{e-}4 \times (8.5)^2 = 1.0206\text{e+}0$; 2.06% deviation at Galactic Center.

Appendix: UQFF Production Framework Reference (v4.75+)

Added by `upgrade_early_whitepapers.py` (v4.75). This appendix cross-references the production physics constants and master equations to enable reproducibility against the current codebase state.

A.1 Calibration Constants

Symbol	Value	Description
$\hat{\Gamma}^0$	$5.0 \times 10^{-6} \text{ day}^{-1}$	UQFF exponential decay rate
[SSq]	0.57	Universal Quantized Factor
$\hat{\Gamma}^2_i$	0.60 ± 0.61	Buoyancy coupling coefficient
k_{DPM}	1.5	Ug1 DPM-dipole coupling
k_{out}	1.2	Ug2 outer-bubble charge coupling
k_{str}	1.8	Ug3 string-rotation coupling
k_{vac}	2.0	Ug4 vacuum-concentration coupling
$\hat{I}$	$10 \text{ kg} \cdot \text{m}^2$	Inertia tensor scale
$E_{\text{react}}(0)$	10 eV	Reference reactive energy

A.2 F_U Master Equation (Complete 4 terms)

$$F_U = U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_{bi} + U_m - \sum_{i=1}^4 \text{igl}[\lambda_i \cdot U_i(r, t) \cdot E_{\text{react}} \text{igr}]$$

Term	Description	Implementation
Ug1	DPM magnetic dipole	compute_Ug1_SOURCE4 / compute_Ug1()
Ug2	Outer-field bubble (charge-reactivity)	compute_Ug2_SOURCE4 / compute_Ug2()
Ug3	Magnetic string rotation	compute_Ug3_SOURCE4 / compute_Ug3()
Ug4	Vacuum concentration (star-BH)	compute_Ug4_SOURCE4 / compute_Ug4()
Ubi	Buoyancy force	compute_Ubi_SOURCE4 / compute_Ubi()
Um	Universal Magnetism (Heaviside-amplified)	compute_Um_SOURCE4 / compute_Um()
$\hat{\Gamma}^0$	4th dissipation term (PAPER_420)	compute_FU_SOURCE4 / full pipeline

4th dissipation term parameters (PAPER_420):

$\hat{\Gamma}^0 = 10^{-6} \text{ day}^{-1}$, $\hat{\Gamma}^2_i = 10 \text{ kg} \cdot \text{m}^2$, $\hat{\Gamma}^2_i = 10 \text{ kg} \cdot \text{m}^2$, $\hat{\Gamma}^2_i = 10 \text{ kg} \cdot \text{m}^2$, $\hat{\Gamma}^2_i = 10 \text{ kg} \cdot \text{m}^2$ (free parameters, not yet empirically calibrated)

A.3 Um Heaviside Phase-Transition Amplifier (PAPER_421)

$$U_m^{\text{full}} = U_m^{\text{base}} \text{imesigl}(1 + 10^{13} \Theta(ho_{SCm} - ho_c) \text{igr}) \text{imesigl}(1 + A_q \cos(\Delta\omega t) \text{igr})$$

Symbol	Value	Description
$\hat{\Gamma}^0_c$	$10 \text{ kg} \cdot \text{m}^3$	SCm critical superconducting density
A_q	0.1	Quasi-periodic beating amplitude (10%)
$\hat{\Gamma}^0$	$2 \times (434 \cdot 365.25) \text{ rad/day}$	434-year Gleisberg supercycle

A.4 UQFF Four Operational Modes

Mode	Dominant Term	Primary Use Case
Compressed	Ug_sum + Newtonian base	Isolated stellar/BH systems

Mode	Dominant Term	Primary Use Case
Resonant	5 resonance frequencies (aDPM, aTHz, $\hat{\alpha}_i$)	Multi-scale field interactions
Buoyant	$\hat{I}_i^2 \tilde{\alpha} Ubi$	Expanding nebulae, stellar winds
Superconductive	$\tilde{\alpha} (1 + 10 \hat{A}^1 \hat{A}^3 \hat{A} \cdot f_H)$	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in MAIN_1_CoAnQi.cpp, CondensedPhysics.py, and CondensedPhysics2.py.